19 Solve algebraically the simultaneous equations

$$2x^2 - y^2 = 17$$
$$x + 2y = 1$$

$x + 2y = 1$, so $x = 1 - 2y$

Substitute: $2(1 - 2y)^2 - y^2 = 17$

1 mark
Eliminate x using the linear equation.

	1	$-2y$
1	1	$-2y$
$-2y$	$-2y$	$4y^2$

$2(1 - 4y + 4y^2) - y^2 = 17$

$2 - 8y + 7y^2 = 17$

1 mark
Expand and substitute correctly.

$7y^2 - 8y - 15 = 0$

1 mark
Correct quadratic.

$a \times c = -105$: $7 \times (-15) = -105$; $7 - 15 = -8$

	$7y$	-15
y	$7y^2$	$-15y$
1	$7y$	-15

$(7y - 15)(y + 1) = 0$

1 mark
Factorise and set each factor to zero.

$7y - 15 = 0$ or $y + 1 = 0$

$y = \frac{15}{7}$ or $y = -1$

$x = 1 - 2 \times \frac{15}{7} = \frac{-23}{7}$ $\quad x = 1 - 2(-1) = 3$

1 mark
Both correctly paired solutions: (−23/7, 15/7) and (3, −1).

(Total for Question 19 is 5 marks)